Demonstration-free Autonomous Reinforcement Learning via Implicit and Bidirectional Curriculum

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Motivation

- Most RL methods assume an episodic setting where **resets to the initial state** are readily available at the end of each episode.
- Autonomous learning of embodied agents involves continual interactions and existing methods often **require costly external interventions** such as human supervision, and scripted policies to simulate resets.
- Recently, Autonomous RL (ARL) framework[1] formalized the non-episodic RL setting but previous works on ARL **rely on demonstration data** to guide the learning process.
- However, a truly autonomous agent should be able to learn from scratch **without external interventions and prior data**.

Contribution

- We propose a demonstration-free ARL algorithm via **Implicit and Bidirectional Curriculum (IBC)** that consists of a conditionally activated **auxiliary agent** and a **bidirectional goal curriculum**.
- To the best of our knowledge, IBC is the first non-episodic RL that can consistently learn **without manual resets and demonstrations** by leveraging curriculum learning.
- IBC achieves **state-of-the-art performance** against previous methods, including even the ones that leverage prior data.

References

[1] Sharma, Archit, et al. "Autonomous Reinforcement Learning: Formalism and Benchmarking." International Conference on Learning Representations. 2021.

[2] Kakade, Sham, and John Langford. "Approximately optimal approximate reinforcement learning." Proceedings of the Nineteenth International Conference on Machine Learning. 2002.

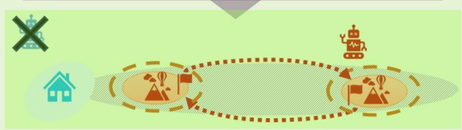
[3] Sharma, Archit, et al. "Autonomous reinforcement learning via subgoal curricula." Advances in Neural Information Processing Systems 34 (2021): 18474-18486.

[4] Sharma, Archit, Rehaan Ahmad, and Chelsea Finn. "A State-Distribution Matching Approach to Non-Episodic Reinforcement Learning." International Conference on Machine Learning. PMLR, 2022.

Overview

- Implicit curriculum

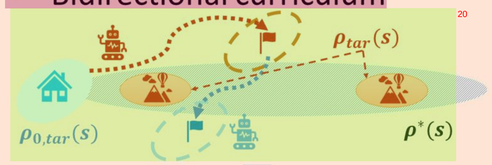
Only forward episode remains at convergence



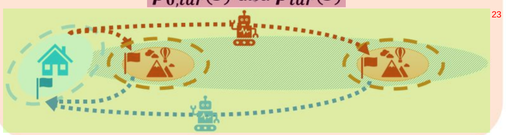
agent of interest (forward agent) : conventional RL agent

auxiliary agent : returns to the initial state

- Bidirectional curriculum



Bidirectional curriculum generation towards $\rho_{0,tar}(s)$ and $\rho_{tar}(s)$

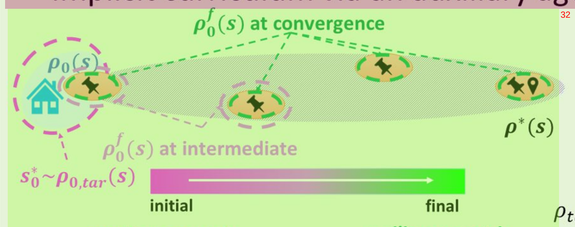


Forward episode curriculum goal

Backward episode curriculum goal

Methods

- Implicit curriculum via an auxiliary agent



auxiliary agent is always activated ($\rho_0^f(s) \to s_0^*$)

auxiliary agent is never activated ($\rho_0^f(s) \to s_{g^*}$)

$\rho_0(s)$: initial state dist.
$\rho^*(s)$: state dist. of π^* that achieves g_{eval}
$\rho_{tar}(s)$, $\rho_{0,tar}(s)$: target goals for agents

: s_{g^*}, a few examples from $\rho^*(s)$
: g_{eval}, goal given during evaluation

$\rho_{tar}(s)$ set to be a few examples from the **state dist. of the optimal policy $\rho^*(s)$** based on the theoretical grounds[2] that $\rho_0(s)$ close to $\rho^*(s)$ enables efficient training

- Bidirectional goal curriculum optimization

$$\max_{\pi} V^{\pi}(T) := E_{(s_0,g)\sim T}[V^{\pi}(s_0,g)]$$

[RL objective]

$$|V^{\pi}(T') - V^{\pi}(T)| \le L \cdot D(T,T')$$

[Lipschitz assumption]

$$\max_{T,\pi}[V^{\pi}(T) - L \cdot D(T,T^*)]$$

[lower-bound form]

original targets $s_{g^*} \sim \rho_{tar}(s)$, $s_0^* \sim \rho_{0,tar}(s)$ for the agents

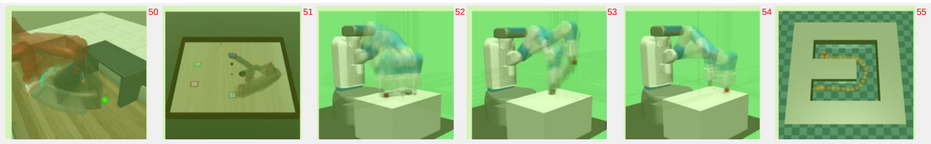
- Curriculum objective

$$\min_{\tau^i=\{s_t^i,\forall t\}\in B} \sum_{(s_0^*,s_{g^*})^i} w\left((s_0^*,s_{g^*})^i,\tau^i\right),$$

where

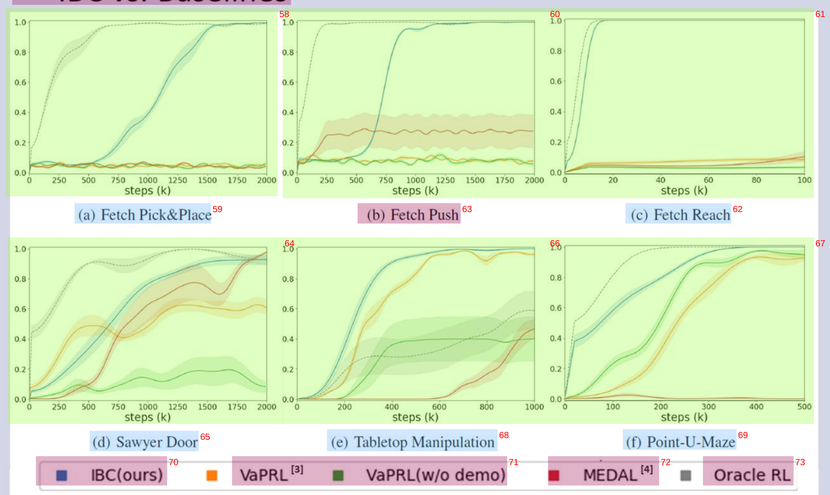
$$w\left((s_0^*,s_{g^*})^i,\tau^i\right) := c\left|\phi_a(s_0^{*,i}) - \phi_a(s_0^i)\right|_2 + \min_t\left(\left|\phi_f(s_{g^*}^i) - \phi_f(s_t^i)\right|_2 - \frac{1}{L}V^{\pi_f}\left(s_0^i,\phi_f(s_t^i)\right)\right)$$

Environments

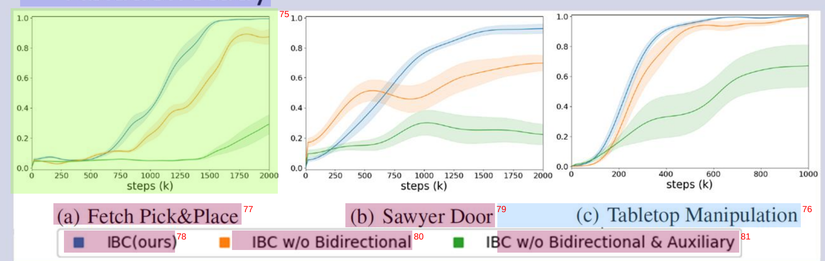


Experiments

- IBC vs. Baselines



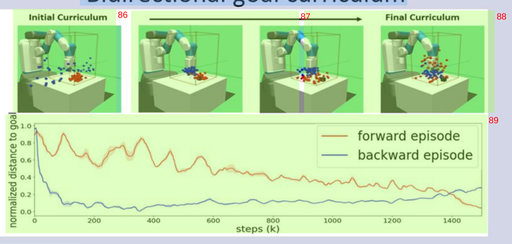
- Ablation study



- Curriculum progress

Implicit curriculum

Bidirectional goal curriculum



- Reward-free experiments